Before class: install the software

Do this before Thursday 12 Nov, 2:00 PM (A3 301). It takes about 20 minutes. In class there is no time to install, and the campus network is slow for large downloads.

You need:

- **NAMD**, an MD program you run from the terminal
- **OpenMM**, an MD library you use from Python (plus numpy and matplotlib for the plots)
- **VMD** (optional), for watching the simulation as a movie

Every step has commands for **Ubuntu** and for **macOS**. On Windows, install WSL (Ubuntu) and follow the Ubuntu commands.

	Ubuntu	macOS
Terminal app	*Terminal* (Ctrl+Alt+T)	*Terminal* (Cmd+Space, type "Terminal")
Shell start-up file	`~/.bashrc`	`~/.zshrc`
How many CPU cores?	`nproc`	`sysctl -n hw.ncpu`
Open an image	`xdg-open file.png`	`open file.png`

Step 1. Unzip the tutorial folder

Unzip `CIS1_Tutorial2_NVE.zip` into `~/Downloads`. You should get a folder `~/Downloads/CIS1_Tutorial2_NVE`. (Any other place works too; then use that path instead of `~/Downloads/CIS1_Tutorial2_NVE` everywhere.)

Step 2. Install NAMD

NAMD is not in the zip because its licence does not allow it to be shared. The NAMD installation guide on Moodle has screenshots.

2a. Download. Register (free) at https://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=NAMD and download the **multicore** build (not CUDA, not "verbs"):

- Ubuntu: `NAMD_..._Linux-x86_64-multicore.tar.gz`
- macOS: `NAMD_..._MacOS...-multicore.tar.gz` (the "universal" build works on both Intel and Apple-silicon Macs)

Either NAMD 2.14 (program name `namd2`) or NAMD 3.x (program name `namd3`) is fine.

2b. Extract it into your home folder and add it to PATH. The `*` matches whichever version you downloaded.

Ubuntu:

```
cd ~/Downloads
tar -xzf NAMD_*_Linux-x86_64-multicore.tar.gz
mv NAMD_*_Linux-x86_64-multicore ~/namd
echo 'export PATH=$HOME/namd:$PATH' >> ~/.bashrc
source ~/.bashrc
```

macOS:

```
cd ~/Downloads
tar -xzf NAMD_*_MacOS*-multicore.tar.gz
mv NAMD_*_MacOS*-multicore ~/namd
xattr -dr com.apple.quarantine ~/namd      # stops macOS blocking an "unverified developer"
echo 'export PATH=$HOME/namd:$PATH' >> ~/.zshrc
source ~/.zshrc
```

2c. Check. One of these should print a help message (both OS):

```
namd2 --help | head -3      # NAMD 2.14
namd3 --help | head -3      # NAMD 3.x
```

Remember which one works. The tutorial writes `namd2`; type `namd3` if that is what you have.

Step 3. Install OpenMM

3a. Conda. If `conda --version` prints a version, skip to 3b. Otherwise install **Miniconda** from https://docs.conda.io/en/latest/miniconda.html:

Ubuntu:

```
cd ~/Downloads
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
bash Miniconda3-latest-Linux-x86_64.sh -b -p $HOME/miniconda3
~/miniconda3/bin/conda init bash
source ~/.bashrc
```

macOS (pick the line for your Mac: Apple menu → *About This Mac* shows "Apple M..." or "Intel"):

```
cd ~/Downloads
curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-arm64.sh    # Apple M1/M2/M3/M4
curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh   # Intel Mac
bash Miniconda3-latest-MacOSX-*.sh -b -p $HOME/miniconda3
~/miniconda3/bin/conda init zsh
source ~/.zshrc
```

3b. Create the tutorial environment (same on both OS):

```
cd ~/Downloads/CIS1_Tutorial2_NVE
conda env create -f environment.yml --solver=libmamba    # creates the environment "cis1-md", about 2 minutes
conda activate cis1-md
```

Without `--solver=libmamba` older conda versions can sit on "Solving environment" for 10 minutes or more. If you see that, press Ctrl+C and run the command above. If you already have a conda environment you want to use, you can install the packages into it instead: `conda install -c conda-forge openmm numpy matplotlib`.

Step 4. Check your install

Same on both OS. Open a **new** terminal first, so the PATH change is loaded:

```
cd ~/Downloads/CIS1_Tutorial2_NVE
conda activate cis1-md
python check_install.py
```

The last line should say `ALL GOOD`. If it does not, find your error below.

If something goes wrong

Message	Fix
`EnvironmentNameNotFound: Could not find conda environment: cis1-md`	The environment was never created. Run Step 3b: `cd ~/Downloads/CIS1_Tutorial2_NVE` then `conda env create -f environment.yml`
`conda: command not found`	Conda is not installed or not initialised: do Step 3a, then open a new terminal
`OpenMM FAILED: No module named 'openmm'`	Wrong environment: `conda activate cis1-md`
`NAMD not on PATH` / `namd2: command not found`	See **Fixing "namd2: command not found"** below
`numpy MISSING` / `matplotlib MISSING`	`conda install -c conda-forge numpy matplotlib`
`Solving environment` takes more than 5 minutes	Ctrl+C, then `conda env create -f environment.yml --solver=libmamba`
`NoWritablePkgsDirError` (during `conda env create`)	Your conda download folder is missing: `CONDA_PKGS_DIRS=$HOME/miniconda3/pkgs conda env create -f environment.yml`
`CondaValueError: prefix already exists`	The environment already exists: just `conda activate cis1-md`
macOS: *"namd2" cannot be opened because the developer cannot be verified*	`xattr -dr com.apple.quarantine ~/namd`, then try again
`Permission denied` when running `namd2`	`chmod +x ~/namd/namd2 ~/namd/charmrun` (use `namd3` if that is yours)

Fixing "namd2: command not found"

The terminal does not know where NAMD is. **1. Find the folder** you extracted (both OS):

```
ls -d ~/namd ~/NAMD_* ~/Downloads/NAMD_* 2>/dev/null
```

Say it printed `/home/you/NAMD_2.14_Linux-x86_64-multicore`. In the commands below, replace `NAMD_2.14_Linux-x86_64-multicore` with **your** folder name (or `namd` if it printed `~/namd`).

2. Add it to PATH permanently, then reload:

Ubuntu:

```
echo 'export PATH=$HOME/NAMD_2.14_Linux-x86_64-multicore:$PATH' >> ~/.bashrc
source ~/.bashrc
```

macOS:

```
echo 'export PATH=$HOME/NAMD_2.14_MacOS-universal-multicore:$PATH' >> ~/.zshrc
source ~/.zshrc
```

3. Check: `namd2 --help | head -3`. For a quick fix in the current terminal only, run the `export PATH=...` part on its own. You can also skip PATH completely and type the full path each time, e.g. `~/NAMD_2.14_Linux-x86_64-multicore/namd2 +p4 nve.namd > out/nve.log`.

Still stuck? Come to class 10 minutes early and we will fix it.

Step 5 (optional). VMD

Download from https://www.ks.uiuc.edu/Research/vmd/ (free registration).

- Ubuntu: extract, then in the extracted folder run `./configure && cd src && sudo make install`. Start it with `vmd`.
- macOS: open the `.dmg` and drag *VMD* to Applications. Start it from Applications (on first launch: right-click → *Open* to get past the security warning).